



Sensitive feet rest on water's surface film, waiting for ripples

Stabbing mouthparts fold away against body when not in use

Fatal vibrations

Hanging upside down at the surface of a pond, a backswimmer uses ripples to find its prey. It ignores powerful splashes, but is attracted by small, rapid ripples—the kind produced when insects fall into the water and struggle to escape. It rows toward their source and attacks its prey from underneath, stabbing it with its tubelike mouthparts.

Backswimmers are aggressive predators and often fight each other for a share of the food.



Synchronized swimmers

When fish swim in a shoal, they move in a coordinated way by using a special pressure-sensing system. Called a lateral line, it keeps these catfish tightly together, making it hard for predators to single out their prey. A fish has two lateral lines—one along each side. Each one consists of a fluid-filled tube connected to the outside by small holes, or pores. Inside the tube is a row of sensors that detect pressure waves. When a fish moves, its neighbors sense the movement with their lateral lines. Within a split second, they move, too.

Skin, hair, and whiskers

Swimming mammals, such as otters, often rely on their sense of touch when they hunt. Otters use their whiskers to find food in murky water and to hunt after dark. Their paws are also very sensitive. They feel for food or hold it down when the otter eats. At sea, seals and walrus also use their whiskers to find food. A walrus has up to 700 whiskers on its upper lip. Together, they form a bristly "moustache" up to 12 in (30 cm) long.

Muzzle contains highest concentration of touch sensors, along with paws

Wired up

Mammals sense touch and pressure with specialized nerve endings, which are set at different depths in their skin. If anything touches the surface of their fur, it makes guard hairs move, which triggers nerves that are wrapped around their roots. Whiskers are even more sensitive. They grow in specialized follicles, and signals from them are processed by a particular part of the brain.

Spreading whiskers sense objects across a wide area

